MAC: XCODE AND QT CREATOR

A compiler you already have, and the IDE we recommend

TWO THINGS, NOT ONE

Compiler vs. IDE, on a Mac

On a Mac, **Xcode** and **Apple Clang** are easy to conflate, but they're two different things:

- **Apple Clang** - the compiler. Installed via the much smaller **Xcode Command Line Tools**, not the full Xcode app.
- **Xcode** - Apple's own IDE. You don't need it for this course - we recommend **Qt Creator** instead, same as Linux and Windows.

INSTALL THEM BOTH

You can use either

Install **XCode** from the App Store, and then get a compiler:

```
xcode-select --install
```

This installs the **Xcode Command Line Tools**, which includes Apple Clang, `make`, and other command-line build essentials.
download required.

RECOMMENDED IDE

Qt Creator on Mac

Free, cross-platform, and reads the same `CMakeLists.txt` every lecture in this course ships with. Install Qt and Qt Creator using the **Qt Online Installer.** selecting just the **Qt Creator** component.

THE DEFAULT KIT

Qt Creator will pick up Apple Clang automatically

Once the Command Line Tools are installed, Qt Creator detects Apple Clang and offers a **Kit** that uses it.

THE CATCH

Apple Clang isn't quite the same as upstream Clang

Apple Clang tracks upstream LLVM/Clang, but on its own release cadence tied to macOS versions - its version numbers don't line up 1:1 with the `clang++ --version` you'd see on Linux. This course is written against **C++23**, which needs a fairly recent one.

> An older macOS with an older Apple Clang may not build every example in this course - that's not a mistake on your part, it's just what "the default compiler" means.

CHECKING YOUR VERSION

Check before you assume

```
clang++ --version
```

If your version is less than 20, you may need to install a newer compiler. **But don't panic** - in a later lecture I will show you how to use **Docker** to get a modern GCC or Clang without touching anything already installed on your system.

OPENING THIS LECTURE

Building this lecture's program

File -> Open File or Project, select a lecture's `CMakeLists.txt`, confirm a Kit, then **Configure Project**. Build and run `rooster` the same way you would any Qt Creator project.